The generator works. The fuel doesn't come.

In a city whose transmission line is repeatedly cut, critical care ends when the diesel ends — not when kW run short. Nobody decides, in advance and with numbers, which loads the last litres should keep alive.

39 days

Gombe–Damaturu–Maiduguri 330 kV line down after tower 372 was vandalised, 22 Sep → 31 Oct 2024 (TCN). Same corridor cut for ~2 months in 2021.

Days dark

Dec 2022: two Maiduguri hospitals without power for days — generators functional, tanks empty (ICIR, 3 Dec 2022).

50%

of hospitals in sub-Saharan Africa report reliable electricity (WHO / World Bank / IRENA / SEforALL, Energizing Health, 2023).

WHO, SPECIFICALLY

- User: works engineer & duty manager of a 250-bed referral hospital in a conflict-affected city
- Beneficiaries: ICU, theatre, maternity, cold-chain & emergency patients — 91,250 bed-days a year
- Payer: hospital management (federal/state budget) — or an energy-service company repaid from diesel no longer bought

ROOT CAUSE

Backup is sized in kW and run on fixed rules. A line outage quadruples diesel need (3,360 L/week normal year, simulated) while deliveries, budgets and convoys stay flat. Status quo at 6,000 L/week (1.8× normal): tank empty 489 h, Tier-1 life-critical circuits dark 487 h of 937.

A fuel-aware brain for hospital microgrids

Digital twin + predictive controller that turns every litre the sun saves into hours of critical care — on the solar, battery and generator a hospital can actually buy.

1 · Fuel-runway governor

Every hour: diesel + battery + forecast solar vs the energy each care tier needs until the next expected delivery. Defers laundry/HVAC/admin first; Tier-1 never. Every decision logged with its arithmetic.

Replay: empty-tank hours 489 → 0

2 · Clinical ride-through reserve

The battery always holds 2 h of Tier-1 while the generator is off. A failed start or a tripped set never darkens the ICU; the battery forms the island with no transfer break.

Replay: Tier-1 breaks 12 → 0; year: 259 → 0

3 · Efficient generator cycling

When it runs, the set runs hard (load + charging) and stops when the battery is full — fewer low-load hours, less wet-stacking, less exposure to running failures.

Year: generator hours 2,667 → 1,381 (−48%)

WHAT MAKES IT NEW

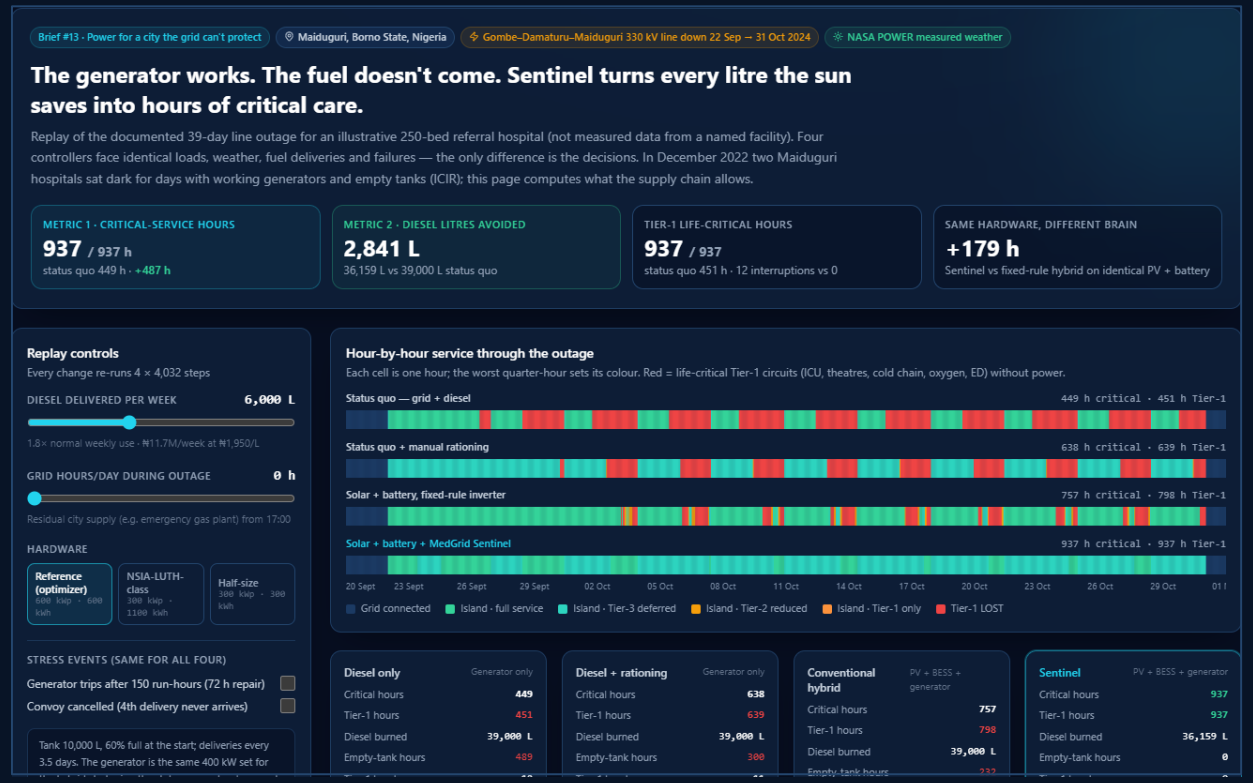
- Treats diesel logistics as the binding constraint — sizing tools (HOMER, REopt) and hybrid inverters don't schedule care tiers against the next convoy
- Isolates the software's value: four controllers on identical hardware, weather, fuel and failures
- Works where hospitals are: offline web app (static, no server) + SMS / USSD alerts on any feature phone

SAME HARDWARE, DIFFERENT BRAIN

+179 h

critical-service hours vs a fixed-rule hybrid inverter on the same 600 kWp + 600 kWh, same 6,000 L/week

Brief metric: “hours of critical service maintained during a line outage, and diesel litres avoided”



Illustrative 250-bed reference hospital (16 tiered circuits) — simulation, not field-measured performance. Real data: NASA POWER hourly irradiance & temperature (Maiduguri 2022–24), NREL/TP-5C00-76553 generator reliability, NERC tariff, Oct-2026 diesel prices.

Same outage · same fuel	Critical h	Tier-1 h	Diesel L	Tier-1 breaks
Status quo (diesel only)	449	451	39,000	12
Diesel + manual rationing	638	639	39,000	11
Solar + battery, fixed rules	757	798	39,000	21
Solar + battery + Sentinel	**937**	**937**	36,159	0

937 islanded hours · 6,000 L/week diesel (1.8× normal use) · 10,000 L tank

449 → 937 h

critical-service hours (Metric 1)

2,841 L

diesel avoided in the outage (Metric 2) · 91,654 L/yr

NEXT OUTAGE · 14-DAY MONTE CARLO (300 trials)

P(Tier-1 kept powered for 14 days), average-reliability generator (NREL data), late/missed deliveries: status quo 0% · fixed-rule hybrid 3% · Sentinel 78%. Well-maintained set: Sentinel 91%.

89 automated tests · power & diesel mass balance every step · NREL '80 % at 2 weeks' reproduced · solar forecast scored on held-out 2024

COST, WHO PAYS, AND THE 6–18 MONTH PATH

Pays back in 3.7 years — or nothing upfront under EaaS

$704k

CapEx (₦1.06B): 600 kWp PV, 600 kWh LFP, 510 kW PCS, Sentinel edge

3.7 yr

payback · IRR 26% · NPV ₦420.5M (20 y @ 18%)

₦311.2M/yr

energy-cost savings on a full measured year (diesel −40%)

₦20.7M/mo

EaaS fee: hospital keeps 20% of savings from month 1; provider IRR 20%

WHO OWNS · PAYS · MAINTAINS

- Energy-service company owns, insures and maintains PV + storage + Sentinel; local O&M technicians; Sentinel subscription $4,800/site-yr is inside the model
- Hospital pays a fixed monthly fee below what it saves — energy cost per bed-day ₦8,940 → ₦8,258; nothing passed to patients
- No grant assumed. Concessional capital widens the margin (NPV at 10%: ₦1.25B)
- Sensitivity: diesel −30% → payback 4.5 yr; CapEx +30% → 4.9 yr; no line outage → 4.1 yr

6–18 MONTHS · WHAT GETS BUILT, BY WHOM

- M1–3: metering + circuit audit at one Maiduguri-class hospital; Sentinel runs in shadow mode on real data
- M4–6: engineering design (IEC 62446 / 62619 / 62109 equipment, NEMSA inspection), EaaS term sheet with a licensed installer
- M7–12: install; Sentinel dispatches; SMS alerts live; measured vs predicted published monthly
- M13–18: sites 2–3 from the fleet screen (Kano, Abuja)
- Biggest risk: real loads differ from the model → shadow-mode gate: proceed only if daily-energy error ≤ ±15%

One hospital, every year — then the fleet

91,654 L
diesel avoided per year (−40%)

246 t
CO_2 avoided per year (2.68 kg/L)

259 → 0
life-critical transfer breaks per year

−1,286 h
generator run-hours per year (−48%)

Portfolio screen
10 site-years simulated (2 × 35,136 steps each)

SITE	GRID REGIME	FACILITY	GHI KWH/M² YR	YIELD KWH/KWP	CAPEX	PAYBACK	NPV 20 Y @18%	IRR	DIESEL AVOIDED	T1 BREAKS AVOIDED/YR	VERDICT
Maiduguri, Borno (NG)	conflict-affected corridor	250-bed referral hospital	2,104	1,545	$704,415	3.7 y	₦420.5M	26.2%	91,654 L (40%)	259	strong
Kano (NG)	typical feeder	250-bed referral hospital	2,127	1,571	$704,415	4.0 y	₦301.6M	23.9%	71,774 L (41%)	300	strong
Abuja (NG)	typical feeder	250-bed referral hospital	1,913	1,428	$704,415	4.5 y	₦159.4M	21.2%	64,492 L (37%)	300	strong
Port Harcourt (NG)	typical feeder	250-bed referral hospital	1,542	1,172	$704,415	5.6 y	₦-86.5M	16.2%	52,795 L (30%)	300	viable
Lagos (NG)	Band A urban	250-bed referral hospital	1,714	1,302	$704,415	6.7 y	₦-254.4M	12.7%	15,740 L (39%)	137	viable
Maiduguri, Borno (NG)	conflict-affected corridor	80-bed general hospital	2,104	1,545	$254,409	3.3 y	₦210.4M	29.3%	40,399 L (41%)	259	strong
Kano (NG)	typical feeder	80-bed general hospital	2,127	1,571	$254,409	3.7 y	₦146.0M	25.9%	31,552 L (42%)	300	strong
Abuja (NG)	typical feeder	80-bed general hospital	1,913	1,428	$254,409	4.2 y	₦91.0M	23.0%	28,557 L (38%)	300	strong
Port Harcourt (NG)	typical feeder	80-bed general hospital	1,542	1,172	$254,409	5.2 y	₦-4.2M	17.8%	23,710 L (32%)	300	viable
Lagos (NG)	Band A urban	80-bed general hospital	1,714	1,302	$254,409	7.2 y	₦-108.7M	11.6%	6,942 L (40%)	137	viable

"Strong" = positive 20-year NPV at an 18 % discount rate and payback < 6 years; "viable" = payback < 10 years. The 80-bed archetype scales loads, PV, storage and generator by 0.35. Lagos uses a Band-A-like regime (fewer, shorter interruptions — an assumption); the Maiduguri row includes the documented 39-day line outage. Prices are national defaults; local diesel premia in insecure regions would raise savings.

Fleet screen: 10 full-year simulations on each site's own NASA POWER weather. Same national prices; differences come from sunshine and grid reliability.

HOW IT TRAVELS

Same engine, new configuration. 10 site-years on measured weather: payback Maiduguri 3.7 y · Kano 4.0 y · Abuja 4.5 y · Port Harcourt 5.6 y · Lagos 6.7 y. Strong in the north, viable in the south — the tool says where it pays. Next: Ghana, Kenya, U.S. (NFPA 99/110) configurations.

TEAM & ASK

Jackson Sixtus Abetianbe — solo entry (product, engineering, finance). We need: one pilot hospital in NE Nigeria, a licensed solar EPC / ESCO partner, 3 months of circuit-level metering, and power-systems + health-facility coaching.